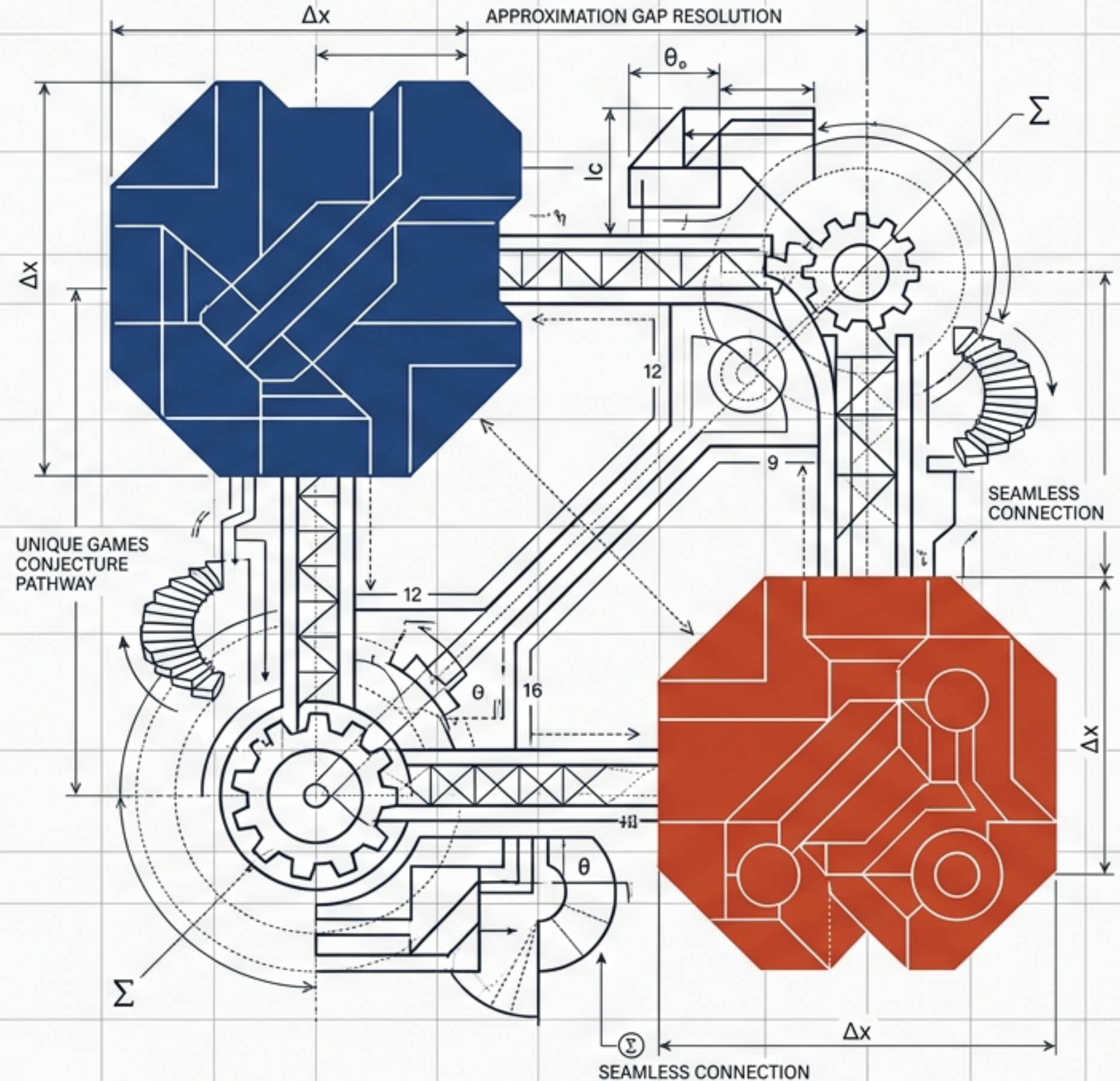
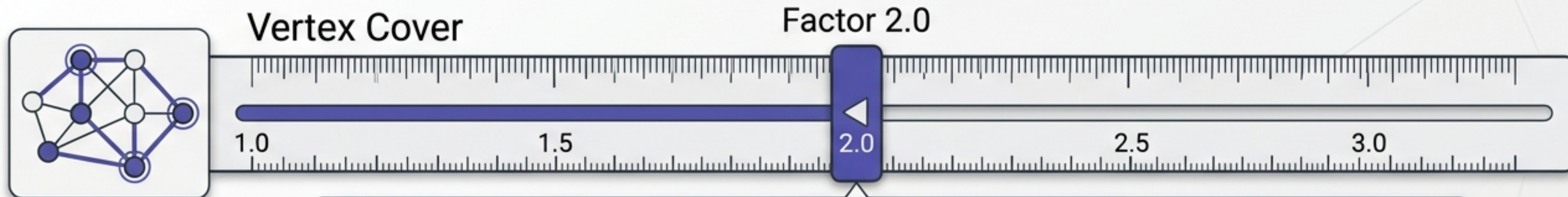


The Blueprint of Complexity

Solving the Unique Games Conjecture and Closing the Approximation Gap



The Limits of Approximation



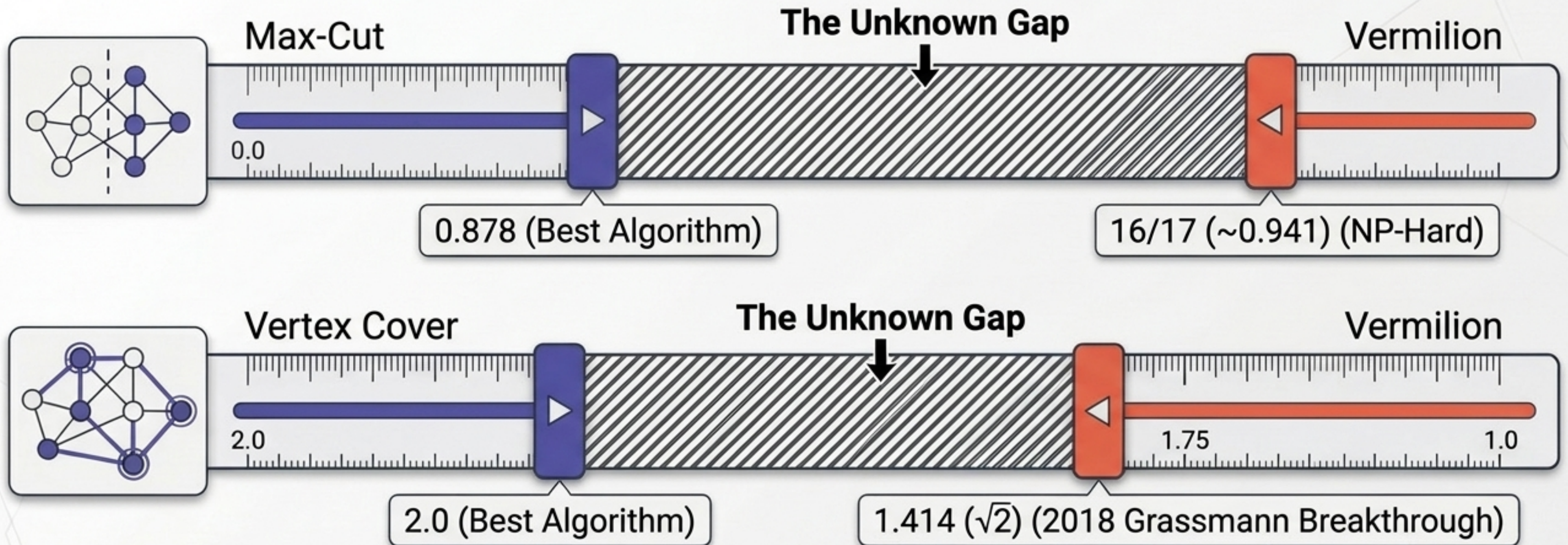
A simple maximal matching algorithm guarantees a factor-2 approximation.



Goemans–Williamson's SDP with a random hyperplane gives ratio 0.878.

Takeaway: We know exact solutions are impossible under $P \neq NP$, but these algorithms established the best known floors and ceilings for approximating NP-hard problems.

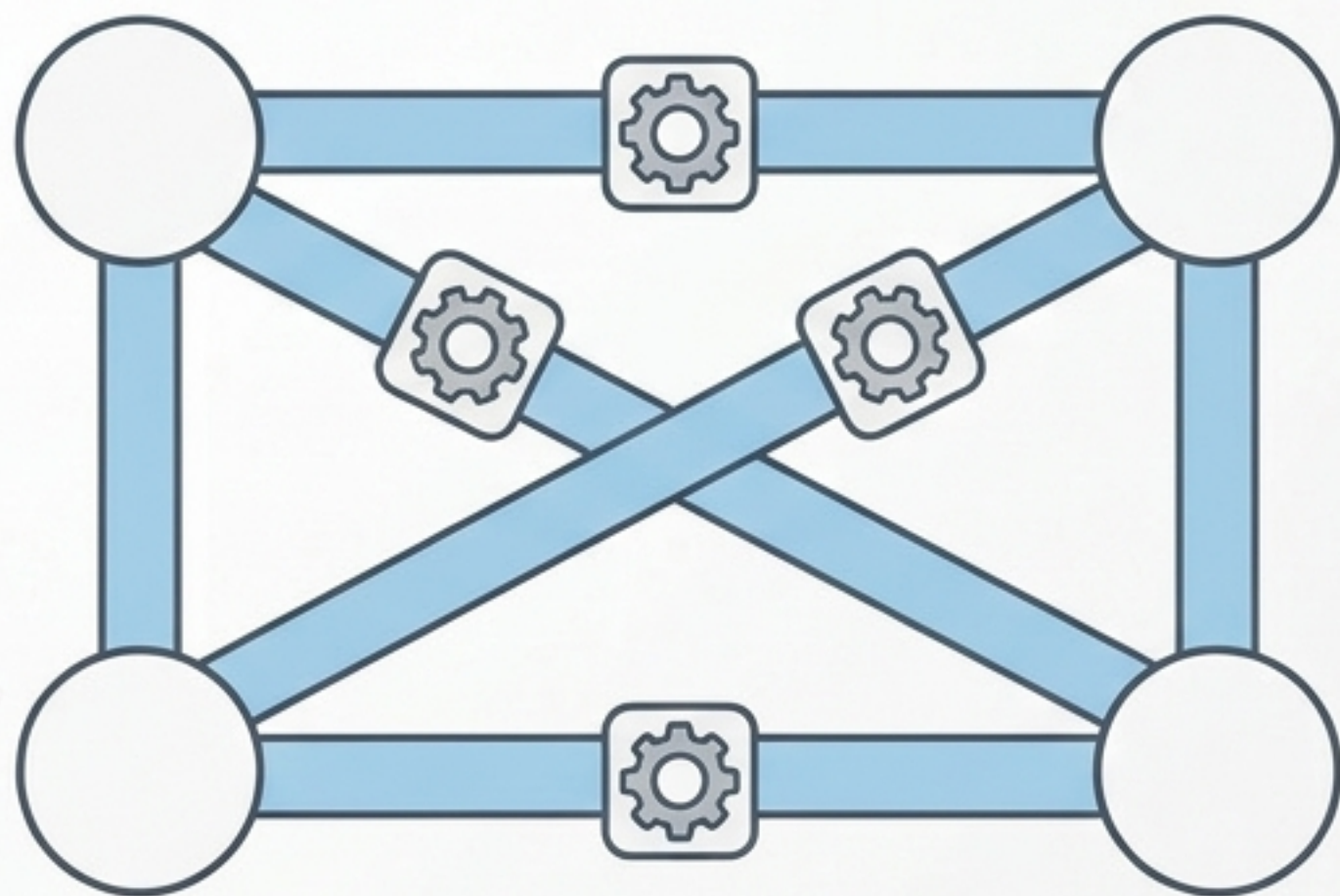
The Historical Void (Pre-2026)



Before 2026, we could not prove if the algorithms were optimal, or if better algorithms simply hadn't been discovered yet. The gap remained mathematically unresolved.

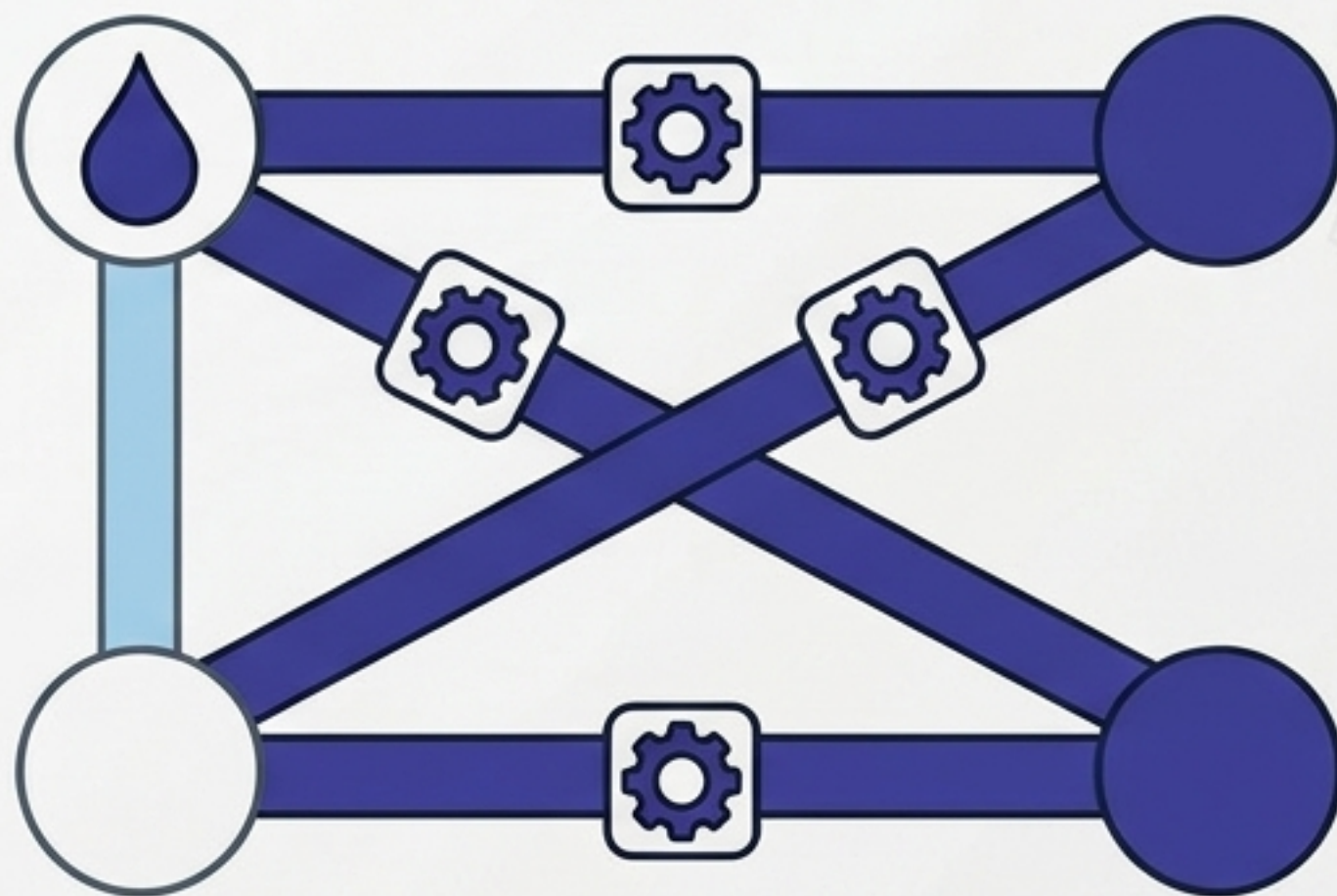
What is a Unique Game?

The Rules



A unique game puts labels on vertices. Each edge rule is a strict 1-to-1 permutation (a color swapper) of the labels.

Perfect Completeness

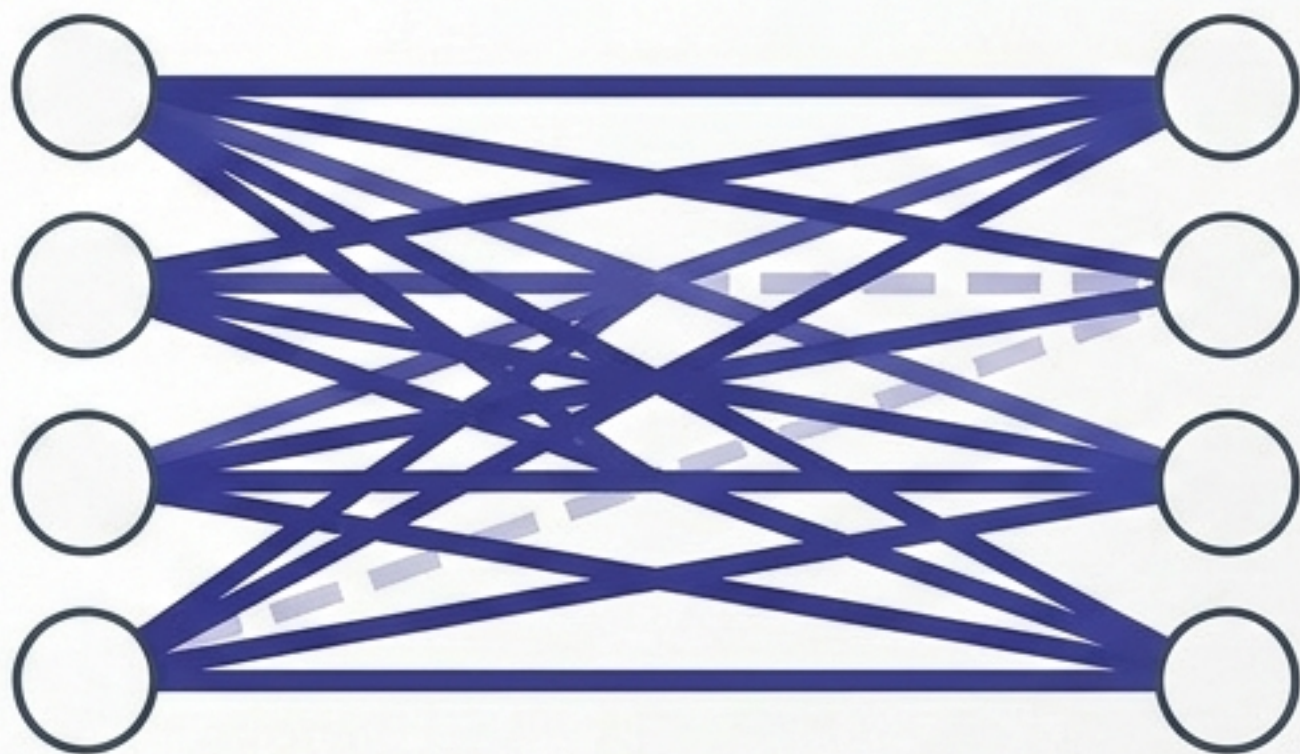


A fully satisfiable unique game is easy to solve: guess one label, and simply propagate it through the graph.

Khot's 2002 Unique Games Conjecture

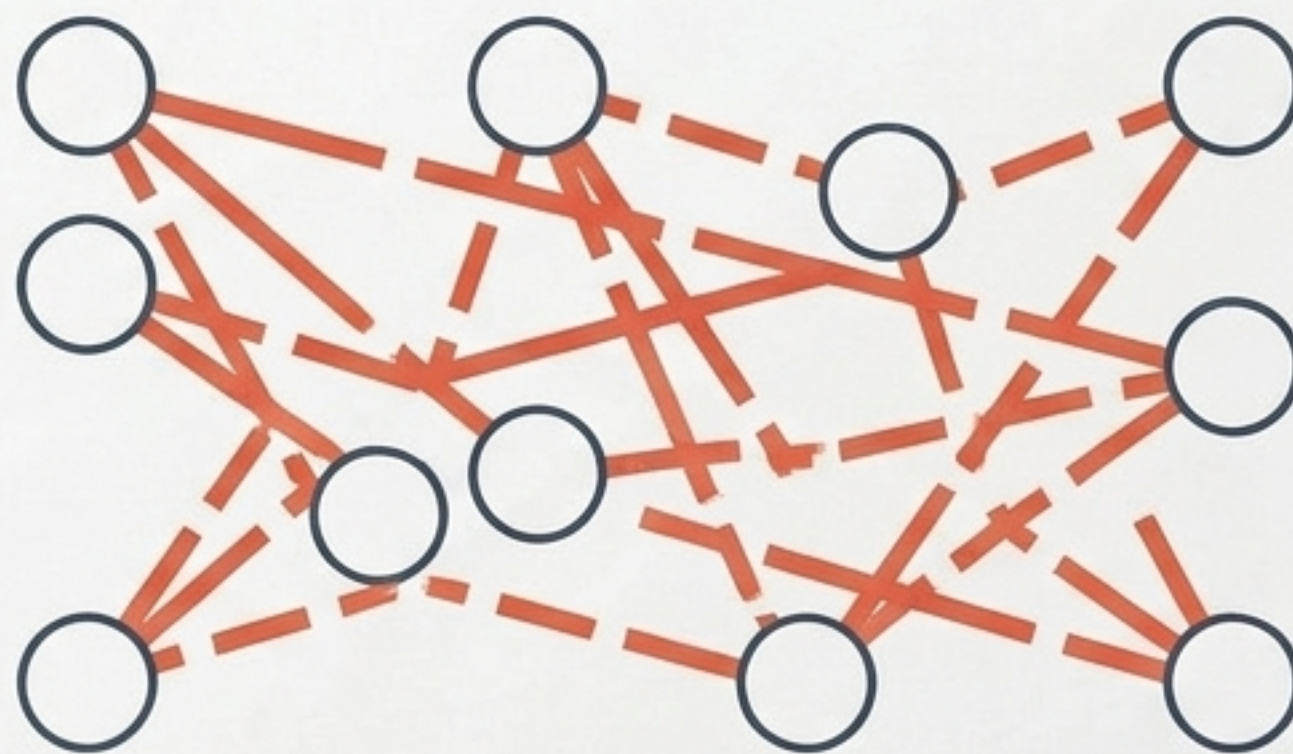
The Khot Gap: NP-Hard to Distinguish

$1 - \epsilon$



Games of value at least $1 - \epsilon$. (Almost fully satisfiable).

δ

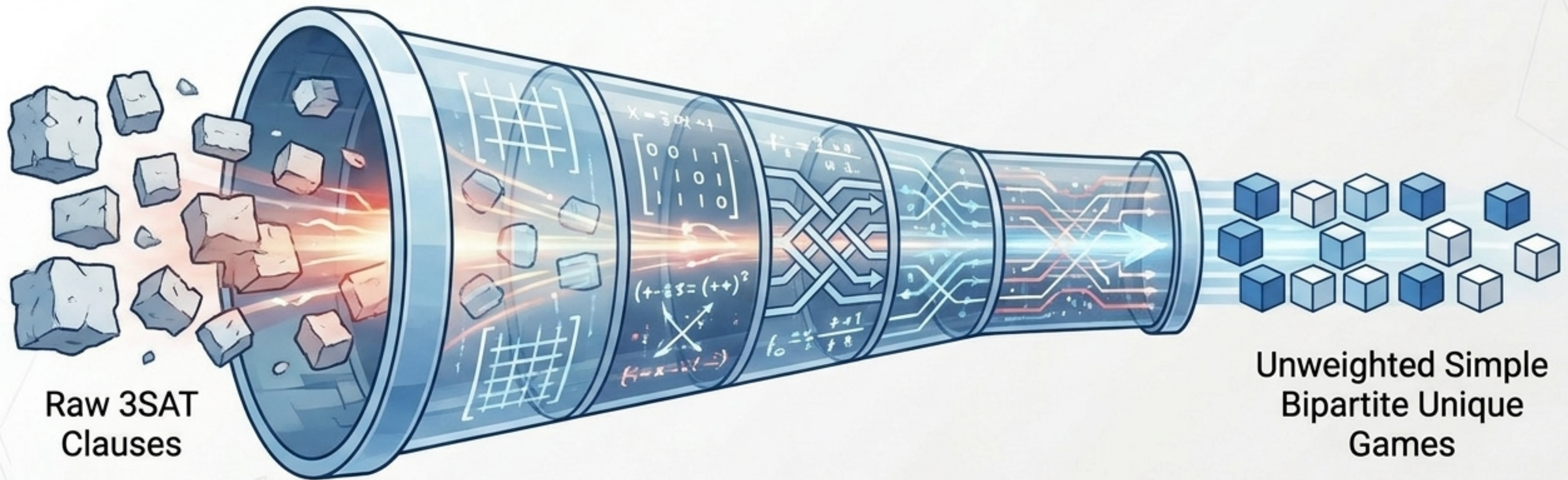


Games of value at most δ . (Pure noise).

Khot conjectured that despite the ease of solving a 100% satisfiable game, telling an almost perfect game apart from a totally broken one is computationally intractable.

[1. 3SAT to Parity] → [2. Non-linear Matrix Decoder] → [3. Fourier/KMS Inverse] → [4. Parallel Repetition]

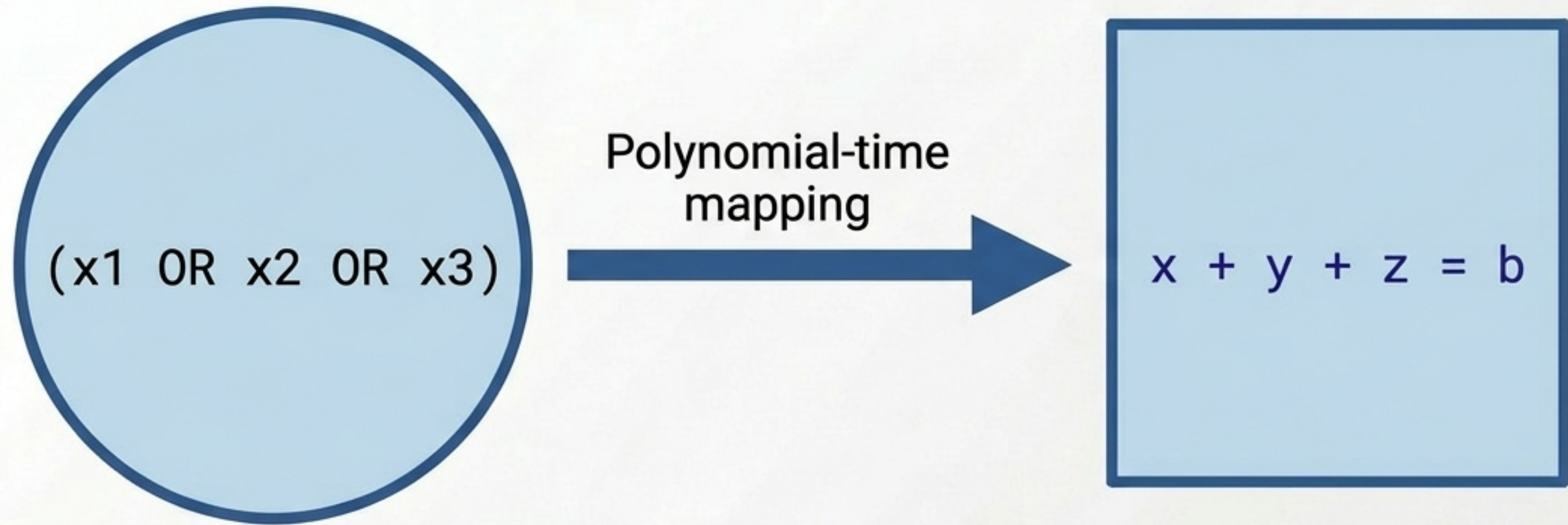
Theorem 1.1: The 2026 Breakthrough



Theorem 1.1 provides a deterministic polynomial-time reduction from 3SAT to unique games over $\mathbb{F}_2^S$, where every constraint is a simple translation.

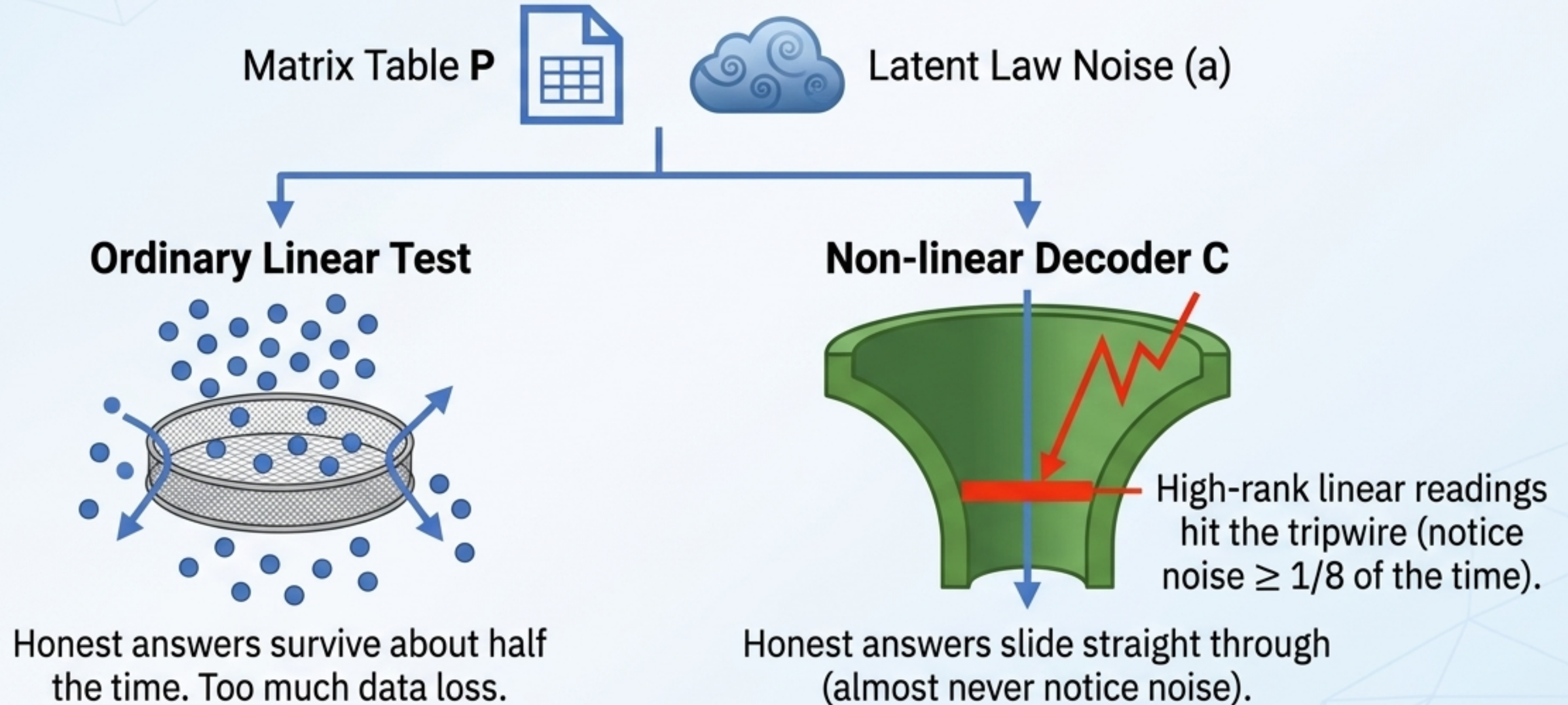
[1. 3SAT to Parity] → [2. Non-linear Matrix Decoder] → [3. Fourier/KMS Inverse] → [4. Parallel Repetition]

Step 1: Håstad's Parity



The proof's foundation doesn't start directly at Unique Games. It begins by mapping 3SAT down into Håstad's parity equations, providing the structured algebraic starting point necessary for the matrix tests that follow.

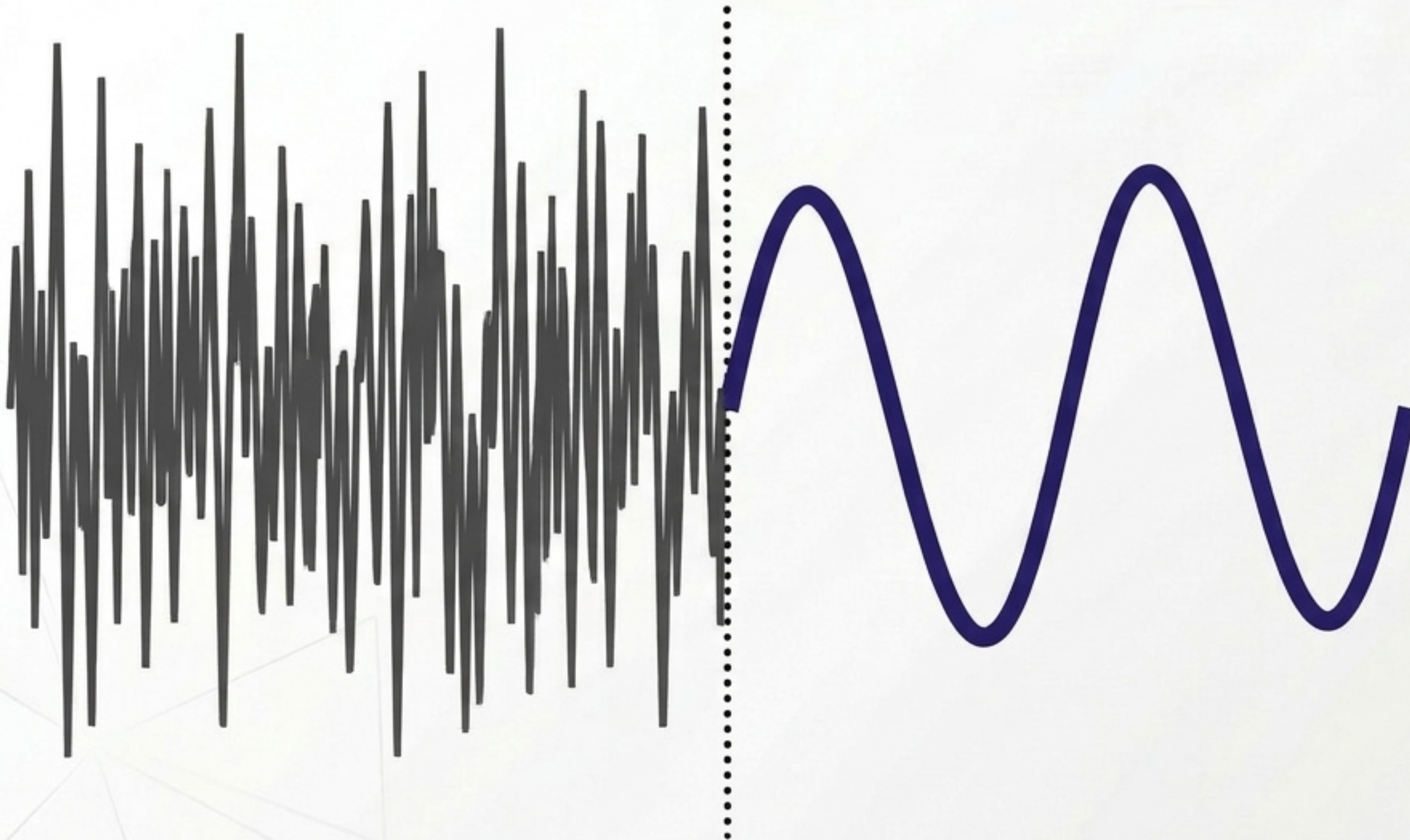
Step 2: The Non-Linear Matrix Filter (The Gadget)



This novel gadget perfectly isolates the signal from the noise, creating the necessary mathematical wedge.

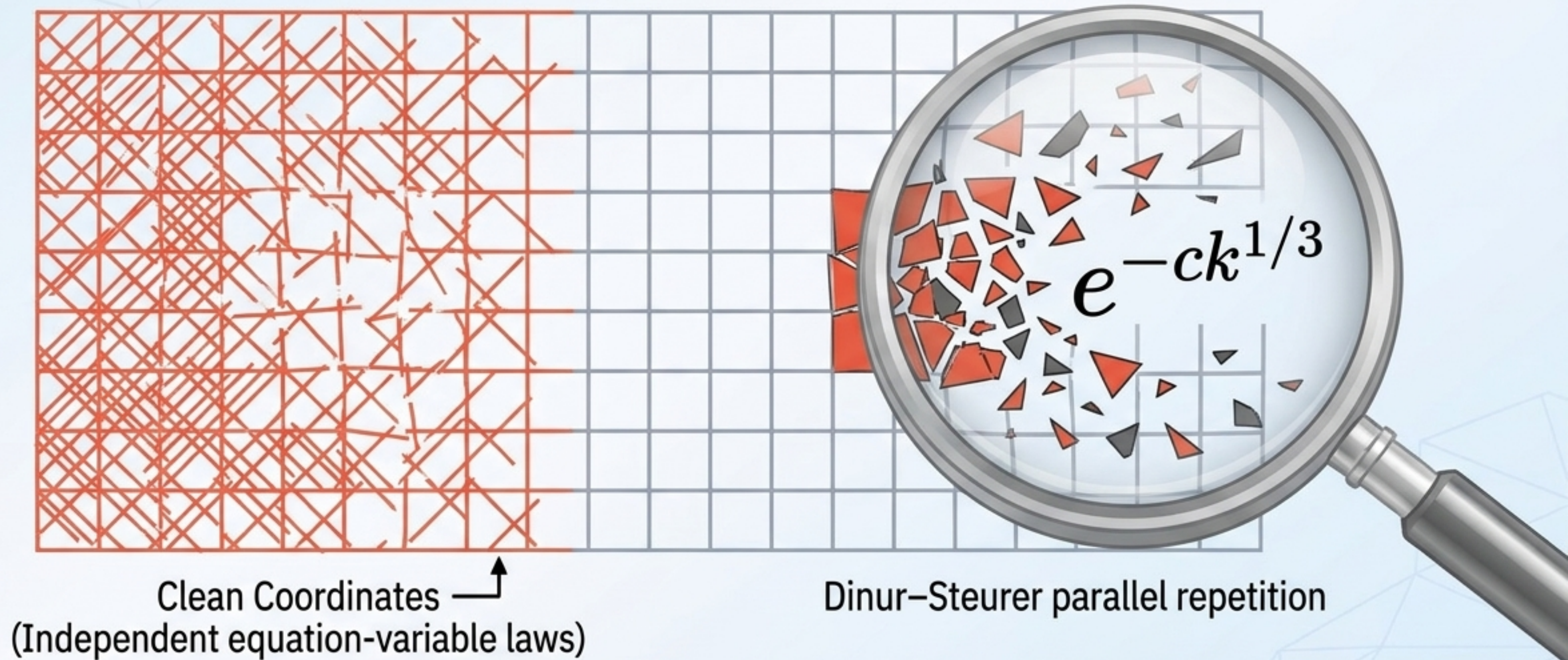
Step 3: Extracting Structure (The YES Case)

Decoder C Boundary



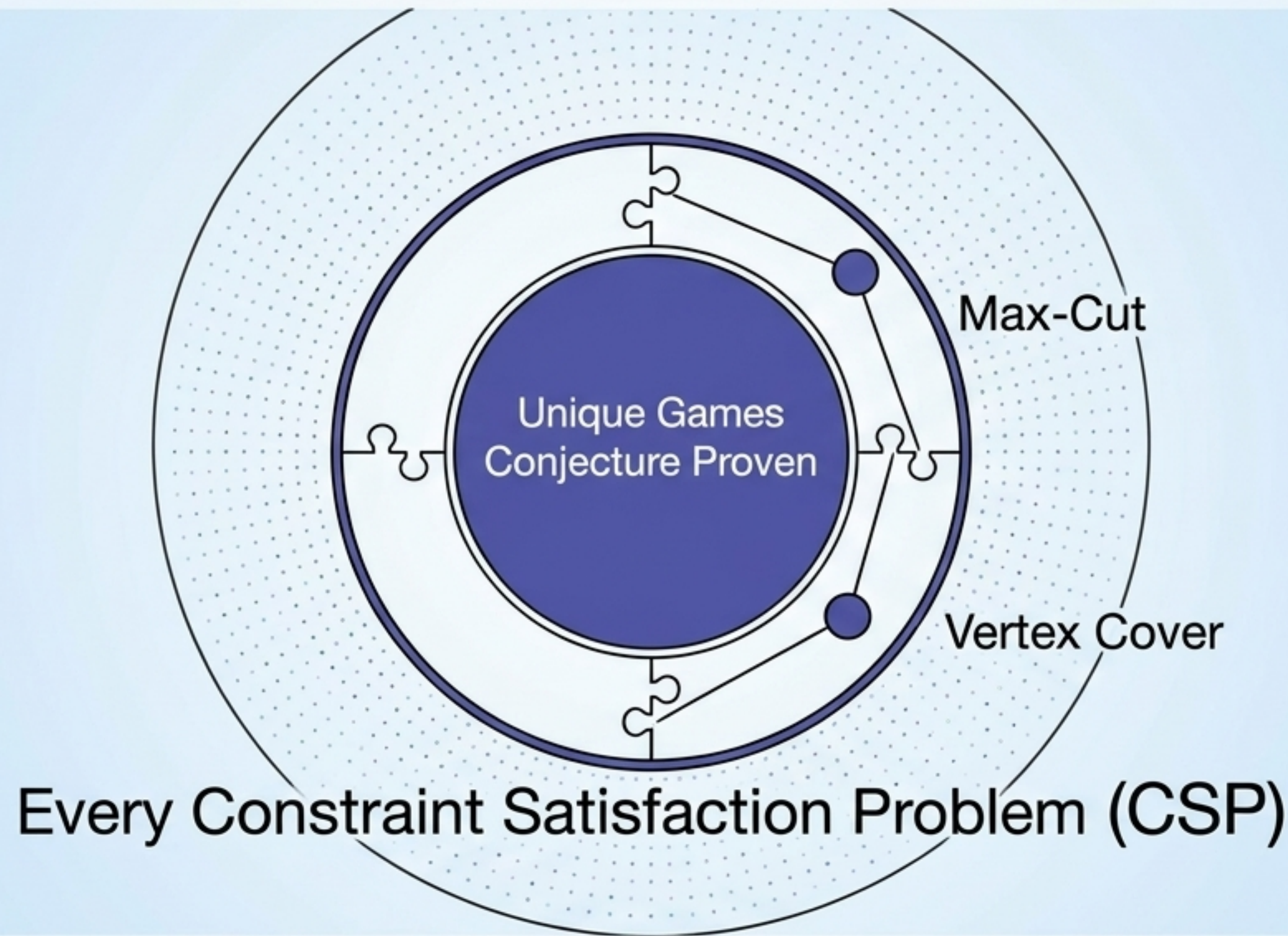
- High acceptance puts most Fourier weight on low rank.
- The Khot–Minzer–Safra (KMS) inverse theorem identifies this heavy low-rank structure.
- Result: Decoded strategies agree with constant probability.

Step 4: Destroying the NO Case



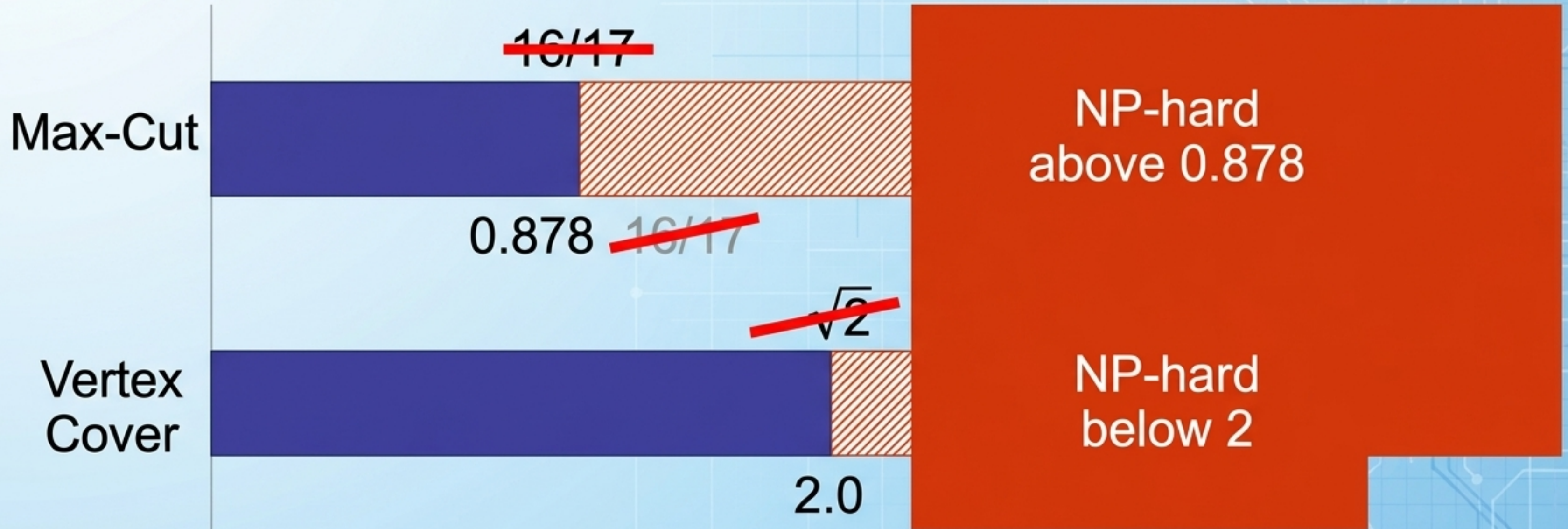
On unsatisfiable inputs, the combination of clean coordinates and parallel repetition makes agreement mathematically microscopic, completely isolating the NO case from the YES case.

The Ripple Effect: Raghavendra's Threshold



Raghavendra's **SDP threshold theorem** stated that if UGC is true, **the exact approximation limits** for every finite CSP are permanently defined. The 2026 proof triggers this universal link.

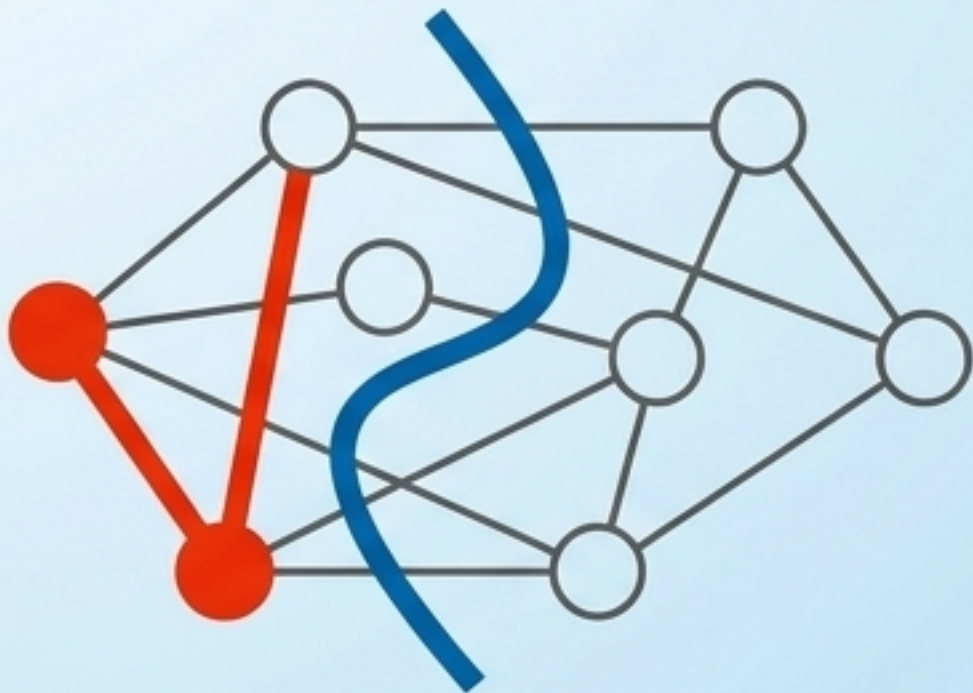
The Gaps Are Closed



Decades of uncertainty erased. Goemans–Williamson's SDP and the basic Factor-2 Matching algorithm are proven absolutely optimal.

Beyond the Classics: New Absolute Limits

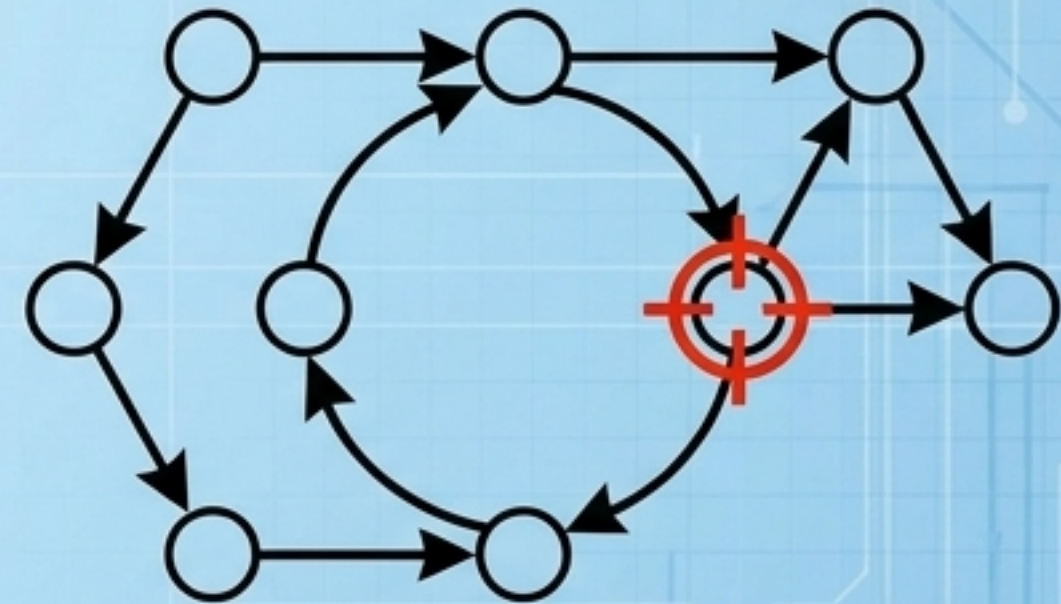
Min-UnCut



Arbitrary constant-factor hardness.

A deterministic reduction constructs instances with arbitrarily large fixed multiplicative gaps.

Directed Feedback Vertex Set



Constant-factor hardness.

Approximating the minimum set to intercept every directed cycle within any fixed constant factor is NP-hard.

The Verified Frontier

Let M be a rank-one matrix. The test accepts if $\exists v \in \mathbb{R}^n$, $\|v\|_2 = 1$ such that $v^T M v \geq \alpha$.

The complexity is $\mathcal{O}(n^2)$ using power iteration

$$\begin{pmatrix} M \\ ca \end{pmatrix} = \begin{pmatrix} m \\ at \end{pmatrix} * \begin{pmatrix} M \\ -b \end{pmatrix} * \begin{pmatrix} vt \\ at \end{pmatrix} \\ = \frac{1}{2} \alpha_i \begin{pmatrix} 1 \\ r_i \end{pmatrix} - a \begin{pmatrix} 1 & v \\ 0 & v \end{pmatrix}.$$

We define the set $S = \{x \mid x \in \{0, 1\}^n \dots\}$ and analyze

$$\sum_{i=1}^n a_i x_i \leq C.$$

```
example (M : Matrix n n ℝ)
  (h : M.rank = 1) : Prop :=
  ∃ (v : n → ℝ), ‖v‖₂ = 1
    ∧ (vᵀ * M * v) ≥ α.
```

implementation uses `Matrix.hasMul`
`Eigenvalues.PowerIteration`.
formalize

```
def S := {x : n → Bool // ...}
```

```
lemma sum_le_C (x ∈ S) :
  ∑ i, a_i * (if x_i then 1 else 0) ≤ C
```

✓
100%
Verified
in Lean

This milestone wasn't just a human triumph. The 2026 result is an AI-generated preprint, and its main theorem is rigorously formalized and verified in the Lean theorem prover via openai/math.

The Final Frontier Remains



The 2026 theorem proves these approximation bounds are definitively NP-hard. But because the result relies on reductions, the ultimate question—whether P equals NP—remains open.